

Title

Prospective evaluation of a selective hepatic LXR antagonist targeting atherogenic lipid phenotypes: study protocol

Abstract

Persistent elevation of circulating lipids despite current therapy remains a significant modifiable risk factor for atherosclerotic disease and cardiac events. The liver X receptor pathway controls both synthesis and disposal of triglyceride-bearing lipoproteins; selective inhibition at this locus may lower disease burden while preserving beneficial cholesterol transport. We describe the protocol for a first-in-human study of TLC-2716, an orally bioavailable compound engineered to restrict LXR antagonism to hepatic and gastrointestinal tissues. The study will assess tolerability, pharmacokinetics, and early biomarker responses in up to 40 healthy or at-risk volunteers over 2 weeks. Measurements will include hepatic enzyme panels, lipid fractionation, insulin secretion, and histologic assessment in liver biopsy organoids. Double-blind, placebo-controlled randomization will occur at a 1:1 ratio, with safety monitoring conducted at predefined intervals.